

PATIENT COMPLEXITY AND PATIENT NEEDS

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The concept of *patient complexity* is intuitive to most primary care clinicians, but difficult to define in an agreed precise manner [1]. While complexity is often narrowly linked to an arbitrary - albeit significant - outcome such as hospital admission risk, predominantly based on medical risk factors [2, 3], clinicians typically experience complexity much more broadly. For them, it manifests as consultations that are multi-faceted, time-consuming, shaped by **psychological, socio-economic and systemic factors**, along with multiple - sometimes problematic - interactions between patient, multiple health care providers and community [4–6]. Some define complexity as the “*gap between an individual’s needs and the capacity of the healthcare system to answer those needs*” [7]. Notably, greater complexity often correlates with higher overall healthcare use and increased social support needs [1, 6].

Several reporting tools for general practice electronic medical records (EMRs) measure patient complexity. This is particularly relevant for reporting to Primary Health Networks (PHNs) as part of their Performance and Quality Framework, which includes indicator ‘P12’ for tracking potentially preventable hospitalisation [8].

- **PenCS** incorporates a CSIRO developed tool using logistic regression to predict hospitalisation using Australian general practice EMR data. Tool workings and development is well-documented and transparent [9], which the authors are willing to share for non-profit purposes. It may be possible to adapt the tool to other outcomes.
- **POLAR** uses the *Diversion 2* tool, developed using machine learning techniques with demographic, diagnoses, medication, pathology and measurement data extracted from general practices with GRHANITE and linked with hospital admission data [10, 11].
- **Primary Sense** integrates the extensively developed, international and proprietary **John Hopkins ACG System** [12, 13]. The John Hopkins ACG system has been used to estimate current and likely future care needs in primary care, including more equitable funding of general practices in the NHS [14, 15]. However, its specific application in Primary Sense is limited to assessing the risk of hospital admission. A core feature of John Hopkins ACG is *patient segmentation*, which groups patients into relatively homogenous groups with similar care needs based on medical and socio-economic characteristics [16–18] (Figure 1).

Opportunities

The concept of **patient segmentation** presents significant potential value in primary care settings. When combined with modifiers such as social need, patient segmentation has the capacity to predict and enable the provision of services tailored towards **behavioural, clinical or demographic drivers of health** over and above predictors based on a single score [16, 18–20] (Figure 2). The patient segmentation approach has proven adaptable for diverse populations, including in Southeast Asia and British Columbia [21, 22] (Figures 3, 4).

Furthermore, developing predictive models for specific outcomes *within* patient segments could **improve predictive power**, as both patient needs and importance of specific risk-factors varies between patient segments [22, 23].

Beyond traditional outcomes like hospital admission, **alternative intermediate outcomes** could be explored to measure complexity and effectiveness. These could include primary care visit frequency and duration, or the volume of correspondence [5].

Figures

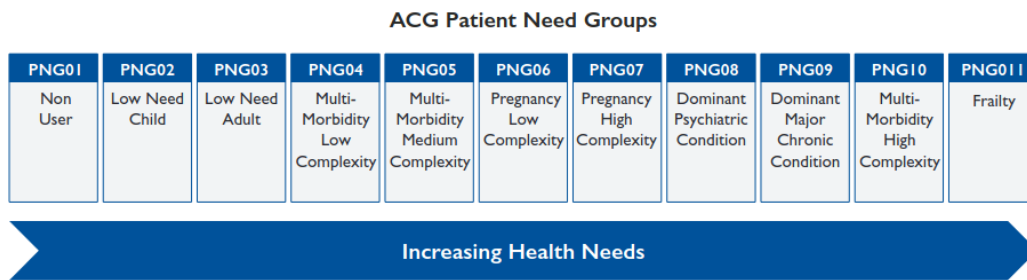


Figure 1: John Hopkins ACG Patient Needs Groups

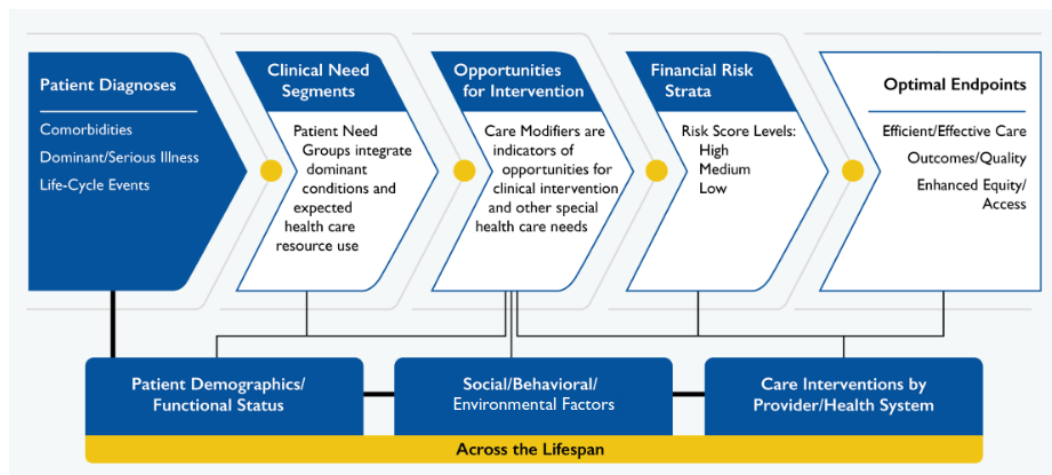


Figure 2: John Hopkins ACG care modifiers

Broad Groups	Children & Adolescence 18 years and below	Adults 19 - 64 years	Seniors 65 years and above
Staying Healthy No Major Health Needs (Preventive Care)	[A] Mostly Healthy Children, Adults and Seniors With no known chronic condition and no known clinical and social risk factor. Maintenance of healthy lifestyle, age appropriate screening, age appropriate preventive health services (e.g. vaccination).		
	[B] At Risk Children and Adults With no known chronic condition and with one clinical or social risk factor such as low socio-economic status and obesity) Targeted screening of at risk population (e.g. genetic/familial diseases), targeted interventions for at risk population.	[C] At Risk Seniors With no chronic condition and one clinical or social risk factor. Interventions to address risk factors to prevent disease.	
Living Well with Illness With Chronic Disease (Chronic Disease Management)	[D] Children with Chronic Condition Individuals who have chronic condition and require minimal follow-up at acute care.	[E] Adults and Seniors with Simple Chronic Condition With condition(s) and case-mix which can be managed primarily in primary care with minimal support from specialist care.	
		[F] Adults and Seniors with Complex Chronic Condition With condition(s) and case-mix which require appropriate co-management between primary and specialist care for chronic disease management.	
		[G] Adults and Seniors with Mental Health Condition With mental health condition. Require primary and specialist care for mental health, with support in the community.	
		[H] Adults and Seniors with Cancer With cancer. Require specialist care with support in the community.	
Getting Well Time-Limited Health Needs (Transitional Care)	[I] Children with Transitional & Long-Term Needs Require additional specialist care to return to baseline function and/or delay progression of disease.	[J] Adults and Seniors with Transitional Needs Had major acute episodic health needs and subsequently require additional support post-discharge and integrate back to the community and return of baseline function.	
Maximising Quality Life Years Frail and End-of-Life (Disease Management and Palliative Care)		[K] Frail Adults and Seniors With diminished function (physical and/or cognitive) or are institutionalised in nursing homes.	
		[L] End-of-Life Children, Adults and Seniors Advanced care planning and accessible end-of-life services to support end-of-life preferences.	

Figure 3: Health Segment Classification of Population Encompassed within Singapore (HealthSCOPES)

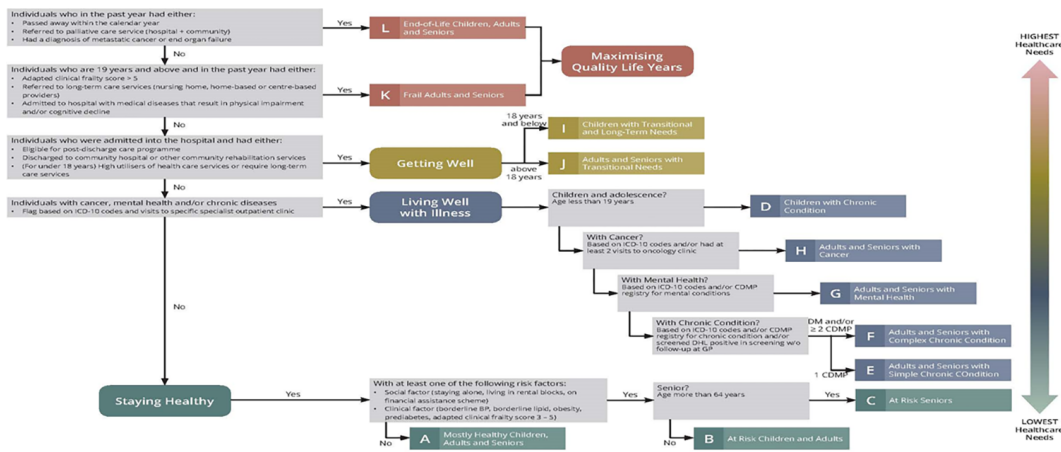


Figure 4: Classification hierarchy of HealthSCOPES, outlining the criteria to categorize an individual into one of the 12 segments

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